

Yaxin Luo

Native Multimodal Systems · Agent Harnesses ·
Long-Horizon Multimodal Tasks

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Education

Aug 2025-Present **MBZUAI** · Ph.D. in Machine Learning Advisor: Dr. Zhiqiang Shen
Sep 2021-Apr 2025 **Technical University of Denmark** · B.Sc. in General Engineering Advisor: Prof. Dimitrios Papadopoulos
Sep 2020-Apr 2021 **University of Edinburgh** · B.Sc. in Mathematics

Research Profile

I build native multimodal AI systems that unify understanding, reasoning, generation, and sustained execution over complex tasks. My current focus is production-grade models for agentic design and automated iteration systems for the long-horizon harnesses that support them.

Research Experience

Meituan · M17 LongCat **Research Intern** Apr 2026-Present

- Research and build Agent Harnesses for **long-horizon multimodal design tasks**, producing trajectories for subsequent model training and integrating execution, tool use, trace capture, automated evaluation, and iterative optimization into a reproducible pipeline.
- Investigate harness self-evolution and meta-optimization: within a fixed model capability boundary, use rollout evidence, evaluator feedback, and component-level updates to improve reliable completion of long-horizon tasks.
- Use **AutoDesign** as Phase I of the long-term LongCat project to validate an end-to-end research loop spanning long-horizon design agents, reusable DesignHarnesses, evaluation protocols, and real deliverables, establishing a systems foundation for subsequent model-and-harness iteration.

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) **Research Assistant** Jan-Aug 2025

- Studied how language-pretrained parameters transfer and generalize to vision-only tasks; developed label-free random-label bridge training for cross-modal parameter adaptation and foundation-capability analysis.
- Built Open CaptchaWorld, an interactive benchmark for multimodal browser agents spanning 20 categories and 225 dynamic CAPTCHAs, quantifying the long-horizon interaction gap between humans and frontier agents.

Open-Source Agent and Harness Systems

AutoDesign **197 Stars · 2026.09.09** 2026-Present

- Built an open-source Design Agent and model-agnostic harness for academic communication artifacts, transforming a paper into an editable poster, slide deck, webpage, and subtitled video; supports coding harnesses including Codex, Claude Code, and Kimi Code.
- Proposed **Meta-Harness Optimization**: an inner Designer-Critic loop iterates native artifacts, while an outer loop updates reusable DesignHarnesses from traces, evaluators, and persistent optimization records, with acceptance gates preventing regressions.
- Built PosterBench over 100 papers. In public results, AutoDesign reached **78.32**, outperforming Claude Design by **7.45** points and delivering **+5.01 to +19.56** point gains across seven matched configurations.

github.com/Yaxin9Luo/AutoDesign

FigMirror **510 Stars · 2026.09.09** 2026-Present

- Contributed to a reference-image-driven scientific figure agent that combines Orchestrator-Drawer-Reviewer roles with Grounded Measurement to turn user data into editable Matplotlib code and camera-ready PDFs.
- Authored early specifications for the method, Web UI, and multi-round refinement; added Codex and Claude runners, event-bus and session-state coverage, and refinement integration tests to establish a verifiable baseline for subsequent agent iterations.

github.com/VILA-Lab/FigMirror

First- and Co-First-Author Publications

01 AutoDesign: Meta-Harness Optimization for Long-Horizon Agentic Design

First author · arXiv 2026 · Long-Horizon Agentic Design

Introduced Meta-Harness Optimization, which recursively improves reusable DesignHarnesses using rollouts, evaluation, and acceptance gates. AutoDesign scored 78.32 on PosterBench, 7.45 points above Claude Design under the same configuration, and was featured by Synced.

arxiv.org/abs/2608.13560 · Synced feature

02 LLMSurgeon: Diagnosing Data Mixture of Large Language Models

First author · ACL 2026 Main · Foundation Model Data Auditing

Formulated pretraining-data mixture diagnosis as Data Mixture Surgery. Recovered latent LLM domain distributions without access to training data using calibrated soft confusion matrices and a constrained inverse problem, and introduced LLMScan for auditable training-recipe evaluation.

aclanthology.org/2026.acl-long.1964

03 Next-Gen CAPTCHAs: Leveraging the Cognitive Gap for Scalable and Diverse GUI-Agent Defense

Co-first author · ICML 2026 · GUI-Agent Defense

Developed a scalable GUI-agent defense framework around the human-agent cognitive gap, designing dynamic tasks that probe differences in interactive perception, memory, decision-making, and action. A data-generation pipeline and backend-supported task types continuously produce diverse CAPTCHA instances at scale.

arxiv.org/abs/2602.09012

04 Language-Pretraining-Induced Bias: A Strong Foundation for General Vision Tasks

First author · TMLR 2026 · Cross-Modal Foundation Models

Introduced label-free random-label bridge training to adapt language-pretrained parameters to foundational vision tasks. Language-induced bias accelerated convergence and improved CIFAR-100 test accuracy by **2.4 percentage points**.

openreview.net/forum?id=N7DSUbnzYo

05 Open CaptchaWorld: A Comprehensive Web-based Platform for Testing and Benchmarking Multimodal LLM Agents

First author · NeurIPS 2025 · Multimodal Agents

Introduced the first dynamic web CAPTCHA benchmark for multimodal LLM agents, covering 20 categories, 225 tasks, and the CAPTCHA Reasoning Depth metric. Humans achieved a 93.3% success rate, compared with 40.0% for the strongest evaluated agent.

arxiv.org/abs/2505.24878

06 γ -MoD: Exploring Mixture-of-Depth Adaptation for Multimodal Large Language Models

First author · ICLR 2025 · Efficient Multimodal Foundation Models

Introduced a Mixture-of-Depth adaptation strategy for existing MLLMs that reduces serving cost through selective token and layer activation while avoiding the substantial degradation caused by directly converting dense layers to MoD.

openreview.net/forum?id=1EnpStvBU8

07 APL: Anchor-based Prompt Learning for One-stage Weakly Supervised Referring Expression Comprehension

First author · ECCV 2024 · Vision-Language Alignment

Designed an Anchor-based Prompt Encoder driven by position, color, and category prompts, together with text-reconstruction and visual-alignment objectives. The method achieved state-of-the-art results on four REC benchmarks, improving over RefCLIP by **6.44%** on RefCOCO.

ecva.net · ECCV 2024 / APL

Collaborative Publications

1. [arXiv 2026](#) · DARTree: Speculative Diffusion Decoding with Autoregressive Draft Trees
2. [arXiv 2026](#) · VideoCoCo: Code-as-CoT for Physically-Consistent Video Generation via an Agentic Dual-Engine System
3. [arXiv 2026](#) · Pushing the Frontier of Black-Box LVLM Attacks via Fine-Grained Detail Targeting
4. [NeurIPS 2025](#) · FADRM: Fast and Accurate Data Residual Matching for Dataset Distillation
5. [ACL 2025](#) · DRAG: Distilling RAG for SLMs from LLMs via Evidence and Graph-based Distillation
6. [CVPR 2025](#) · DViN: Dynamic Visual Routing Network for Weakly Supervised Referring Expression Comprehension
7. [arXiv 2025](#) · Dynamic Pyramid Network for Efficient Multimodal Large Language Model
8. [arXiv 2025](#) · Dataset Distillation via Committee Voting

Additional Experience

- IEEE Cybermatics Congress 2024: volunteer organizing-team member; served as conference assistant and session chair for the Smart Data workshop.
- SciSec 2024: conference volunteer. Institute of Social Science Survey, Peking University: Summer Intern, Public Mental Health Project (Jul-Sep 2019).